DANIEL THOMAS

MATERIALS & METALLURGICAL ENGINEER

Materials Science | Materials Informatics | Computational Materials
Artificial Intelligence | Machine Learning | Sustainable Materials

Engineering sustainable materials through materials science, computation, data and artificial intelligence.
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PROFESSIONAL SUMMARY

I am a Materials and Metallurgical Engineering graduate with a strong academic foundation and growing expertise in materials science, materials informatics, computational materials science, machine learning and sustainable materials development. Interested in combining experimental and data-driven methods to design advanced materials and optimize production processes. Aspires to pursue a Master's degree in Materials Science and Engineering, contribute to innovative research and become a professor focused on sustainable materials and AI-driven engineering.

PROFESSIONAL EXPERIENCE

Research & Academic Projects 2021–2026

- Developed an eco-friendly epoxy hybrid composite reinforced with bio-waste for automotive application as a final-year project.
- Built engineering software and analysis systems using Python, including composite prediction, limestone extraction analysis, grinder and polishing machine software, and characterization analysis.
- Developed a materials informatics program to predict material properties using experimental data and AI models.

Teaching / Volunteering

- Tutored students in physics, chemistry and mathematics at primary, junior-secondary and university levels.
- Taught engineering mathematics, heat and mass transfer, transfer functions, ionic equilibrium diagrams, Fourier series and Laplace transforms.
- Helped people manage applications and built websites and applications, including a fellowship registration database using Supabase.

Entrepreneurship Centre 2023–2024

Federal University Entrepreneurship Centre

- Instructed participants in soap, cream and related product preparation.

Industrial Training – NITDA Jul–Dec 2025

- Gained practical exposure to data management, cloud computing (AWS, Azure and Google Cloud), Python programming, web development, database management and system deployment.

Mineral Processing Laboratory Sep–Nov 2026

- Practical exposure to bauxite and mineral-processing activities, jaw crushing, sorting equipment, testing machines, casting, corrosion studies, fabrication and analysis.

Nanotechnology Research Group 2024

- Attended seminars and gained exposure to nanotechnology and advanced-materials research; received a certificate of participation.

EDUCATION

Federal University of Technology, Minna 2021–2026

B.Eng. Materials and Metallurgical Engineering
Department of Materials and Metallurgical Engineering
CGPA: 4.33

SKILLS

PROGRAMMING	ENGINEERING
• Python • Flask, JavaScript (basic) • HTML, CSS, Tailwind CSS • Git & GitHub • Supabase • VS Code	• Materials Science • Metallurgical Engineering • Composite Materials • Materials Characterization • Mechanical Testing • Sustainable / Automotive Materials
COMPUTER & DATA	**SOFT SKILLS**
• Data Analysis • Data Visualization • Machine Learning • Computational Materials Science • Materials Informatics • Scientific Computing	• Problem Solving • Critical Thinking • Teamwork • Communication • Adaptability • Time Management

LANGUAGES

- Igbo
- English – Fluent
- German – Basic (learning)

MEMBERSHIPS & CERTIFICATIONS

- Nanotechnology Research Group – Member
- Computer Ethics, IT Ethics and Professionalism in Computing – Certificate & ICDL
- Herbalism / plant-based product preparation – Training or certificate & FUTECH hub

PROFESSIONAL STRENGTHS

- Research and innovation
- Data-driven problem solving
- Materials and composite development
- Programming and software development
- Teaching and mentorship
- Leadership and teamwork

CAREER GOAL

To become a professor and contribute to developing new, sustainable materials while using AI to optimize production processes and engineering solutions.

RESEARCH & FUTURE DIRECTION

- Master's in Materials Science / Materials Engineering
- Materials Informatics & Machine Learning
- Computational Materials Science
- Sustainable Materials
- Advanced Composite Materials
- AI-assisted Materials Discovery

SECONDARY EDUCATION

NEPA Staff Secondary School 2006–2012

W.A.E.C -Science Student

AWARDS & RECOGNITION

- Best Graduating Student – B.Eng. (FUT Minna)
- Excellent performance in Geography (Secondary School)
- Excellent performance in Mathematics (Secondary School)

MANAGERIAL, AND SOCIAL SKILLS

• Strategic Planning and Coordination • Team Building and Collaboration • Problem-Solving and Troubleshooting • Adaptability and Flexibility • Goal Setting and Target Achievement • Project and Activity Coordination • Delegation and Accountability •Initiative and Independent Working • Integrity and Professional Ethics & Attention to Detail • Creativity and Innovation

HOBBIES & INTERESTS

- Reading and independent study
- Biblical studies and spiritual development
- Solving mathematical problems
- Playing musical instruments, especially the keyboard
- Weightlifting and strength training
- Exploring scientific and technological developments



LEADERSHIP & RESPONSIBILITIES

•**Vice President, Department of Materials and Metallurgical Engineering:** Federal University of Technology, Minna. Provided student leadership and supported departmental activities. Organised online programs for fellow students via zoom 2024

•**ELCOM Chairman:** Coordinated electoral activities and oversaw the conduct of elections for student political positions as chairman.2026

•**Academic Group Leader:** Led student groups during academic assignments and practical sessions, coordinating teamwork and task completion.2022 to 2026

•**Classroom Teacher and Acting Head Teacher:** Supported teaching, classroom administration, student supervision, and the maintenance of an orderly learning environment.2015 -2020

• **Class Monitor & Academic Lead (**High School**)** Served as the top-ranked student and academic representative, coordinating class activities, maintaining an orderly learning environment, and serving as the liaison between students and teachers.2006-2007